



6561 - Legacy Observations of BHXRBs: wind driving mechanism and relation with the jet

Cycle: 2, Proposal Category: GO

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				
	2		MIRI Low Resolution Spectroscopy	(1) BHXRB-transient
	3		MIRI Low Resolution Spectroscopy	(1) BHXRB-transient

ABSTRACT

Edge-on black hole transients show blue-shifted absorption lines in their X-ray spectra due to hot equatorial disk winds. Blue-shifted absorption lines also been discovered in optical and UV. These features must be produced in an outflow, but the physical conditions required to form them are very different. The three features have never been observed at the same time. It is unclear if they are linked to distinct outflows or simply to different regions. We propose to answer this question with simultaneous time-resolved spectroscopy of a transient BHXRB in X-ray, UV, optical and mid-IR during their hard- intermediate- and soft-states. This will allow us to test if all types of wind features are present simultaneously and whether they display correlated variability among them and with the jet

OBSERVING DESCRIPTION

For JWST, we request 2×3h with low-resolution spectroscopy with MIRI (P750L disperser), of which more than 2h would be overhead giving an effective exposure time of 50 min (11 dithers with 100). We will reach a S/N 100 10 across all full spectral range (5-14 μ m, respectively) for flat spectrum at $F = 20$ mJy.

Proposal 6561 - Targets - Legacy Observations of BHXRBs: wind driving mechanism and relation with the jet

Generic Targets	#	Name	Criteria	Description
	(1)	BHXRB-transient	Black Hole Transient	

Proposal 6561 - Observation 2 - Legacy Observations of BHXRBS: wind driving mechanism and relation with the jet

Observation	Proposal 6561, Observation 2 Tue Dec 19 14:00:15 GMT 2023 Diagnostic Status: Error Observing Template: MIRI Low Resolution Spectroscopy							
Diagnostics	(Observation 2) Error (Form): Acq Groups/Int is a required field. (Observation 2) Error (Form): AcqFilter is a required field. (Observation 2) Error (Form): Dither Type is a required field. (Observation 2) Error (Form): Groups/Int is a required field. (Observation 2) Error (Form): Observations using Generic targets must have an On Hold or Target Of Opportunity special requirement. (Observation 2) Error (Form): Only 4, 6, 8, 10, 12, 22, 36, 44, 66, 86 or 98 are allowed for Groups/Int. (Observation 2) Error (Form): Subarray is a required field. (Exposure) Error (Form): Acq Groups/Int is a required field. (Exposure) Error (Form): Only 4, 6, 8, 10, 12, 22, 36, 44, 66, 86 or 98 are allowed for Groups/Int. (Observation 2) Warning (Form): Record ETC Wkbk.Calc ID used to verify target acquisition. (Exposure) Warning (Form): Record ETC Wkbk.Calc ID used to verify target acquisition. (Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.							
Generic Targets	#	Name	Criteria	Description				
	(1)	BHXRb-transient	Black Hole Transient					
Acquisition	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time ETC Wkbk.Calc ID
	1	SAME		FAST		1	1	0.0
Template	Subarray			Obtain Verification Image?				
				false				
Dithers	#	Dither Type	No. Spectral Steps	Spectral Step Offset	No. Spatial Steps	Spatial Step Offset		
	1							
Spectral Elements	#	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Exposures/Dith	Total Dithers	Total Exposure Time ETC Wkbk.Calc ID
	1	FASTR1		1	1	1	1	0.0

Proposal 6561 - Observation 3 - Legacy Observations of BHXRBs: wind driving mechanism and relation with the jet

Observation	Proposal 6561, Observation 3 Tue Dec 19 14:00:15 GMT 2023 Diagnostic Status: Error Observing Template: MIRI Low Resolution Spectroscopy								
Diagnostics	(Observation 3) Error (Form): Acq Groups/Int is a required field. (Observation 3) Error (Form): AcqFilter is a required field. (Observation 3) Error (Form): Dither Type is a required field. (Observation 3) Error (Form): Groups/Int is a required field. (Observation 3) Error (Form): Observations using Generic targets must have an On Hold or Target Of Opportunity special requirement. (Observation 3) Error (Form): Only 4, 6, 8, 10, 12, 22, 36, 44, 66, 86 or 98 are allowed for Groups/Int. (Observation 3) Error (Form): Subarray is a required field. (Exposure) Error (Form): Acq Groups/Int is a required field. (Exposure) Error (Form): Only 4, 6, 8, 10, 12, 22, 36, 44, 66, 86 or 98 are allowed for Groups/Int. (Observation 3) Warning (Form): Record ETC Wkbk.Calc ID used to verify target acquisition. (Exposure) Warning (Form): Record ETC Wkbk.Calc ID used to verify target acquisition. (Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.								
Generic Targets	#	Name	Criteria	Description					
	(1)	BHXRb-transient	Black Hole Transient						
Acquisition	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	SAME		FAST		1	1	0.0	
Template	Subarray				Obtain Verification Image?				
					false				
Dithers	#	Dither Type		No. Spectral Steps	Spectral Step Offset		No. Spatial Steps		Spatial Step Offset
	1								
Spectral Elements	#	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Exposures/Dith	Total Dithers	Total Exposure Time	ETC Wkbk.Calc ID
	1	FASTR1		1	1	1	1	0.0	